



Clinical manifestations, diagnosis, and prognosis of Crohn disease in adults

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INTRODUCTION

Inflammatory bowel disease (IBD) is comprised of two major disorders: ulcerative colitis (UC) and Crohn disease (CD). UC affects the colon and is characterized by inflammation of the mucosal layer. CD is characterized by transmural inflammation and may involve any portion of luminal gastrointestinal tract, from the oral cavity to the perianal area. Patterns of CD distribution include the following:

- Approximately 80 percent of patients have small bowel involvement, usually in the distal ileum, with one-third of patients having ileitis exclusively.
- Approximately 50 percent of patients have ileocolitis, which refers to involvement of both the ileum and colon.
- Approximately 20 percent have disease limited to the colon. In contrast to rectal involvement in patients with UC, one-half of CD patients with colitis have sparing of the rectum.
- Approximately one-third of patients have perianal disease (eg, perianal fistula).
- Approximately 5 to 15 percent have involvement of the mouth or gastroduodenal area, while fewer patients have involvement of the esophagus and proximal small bowel.

CD can be classified based on age of onset, disease location, and disease behavior (eg, stricturing), and the Montreal classification is commonly used for epidemiologic and

population-based studies ([table 1](#)) [1]. (See "[Definitions, epidemiology, and risk factors for inflammatory bowel disease](#)".)

This topic will review the clinical manifestations, diagnosis, and prognosis of CD in adults. The pathogenesis of IBD is discussed separately. (See "[Immune and microbial mechanisms in the pathogenesis of inflammatory bowel disease](#)".)

Management of patients with CD including assessment of disease activity, severity and risk, is discussed separately:

- (See "[Overview of the medical management of mild \(low risk\) Crohn disease in adults](#)".)
- (See "[Medical management of moderate to severe Crohn disease in adults](#)".)
- (See "[Management of Crohn disease after surgical resection](#)".)
- (See "[Perianal Crohn disease](#)".)
- (See "[Surveillance and management of dysplasia in patients with inflammatory bowel disease](#)".)

CLINICAL FEATURES

Patient presentation — Patients with CD can have symptoms for many years prior to diagnosis or they may present acutely [[2,3](#)].

Cardinal symptoms — The cardinal symptoms of CD include abdominal pain, diarrhea (with or without gross bleeding), fatigue, and weight loss [[4](#)]:

- **Abdominal pain** – Crampy abdominal pain is a common manifestation of CD, regardless of disease distribution:
 - A patient with disease limited to the distal ileum frequently presents with right lower quadrant abdominal pain.
 - The transmural nature of the inflammatory process can over time result in fibrotic strictures. These strictures often lead to repeated episodes of abdominal pain and small bowel obstruction, or less commonly colonic obstruction. If the inflammation is aggressive, the transmural nature results in a fistula with abscess formation.
 - Some patients have no symptoms of CD until luminal narrowing causes abdominal pain and early signs of obstruction (fewer bowel movements).
- **Diarrhea** – Diarrhea is a common presentation, but bowel symptoms often fluctuate over a long period of time [[5](#)]. A history of persistent but intermittent diarrhea without

gross blood and with other features of IBD (eg, skin, eye, or joint problems) suggests the diagnosis of CD. Diarrhea associated with CD may have multiple causes including:

- Excessive fluid secretion and impaired fluid absorption by inflamed small or large bowel
- Bile salt malabsorption due to an inflamed terminal ileum
- Steatorrhea related to loss of bile salts
- Enteroenteric fistulas causing bypass of portions of absorptive surface area

Stools frequently contain microscopic blood (eg, positive guaiac or immunochemical test); however, some patients with CD with predominantly colonic involvement may have grossly bloody stools.

- **Systemic symptoms** – Fatigue is a common feature of CD. Weight loss is often related to decreased oral intake because patients with obstructing segments of bowel feel better when they do not eat. Weight loss may also be related to malabsorption.

Fever occurs less frequently and may be due to the inflammatory process itself or may be the result of intestinal perforation complicated by an intra-abdominal abscess. (See ['Features of transmural inflammation'](#) below.)

Features of transmural inflammation — Transmural bowel inflammation of CD is associated with sinus tracts that may lead to fistulas and inflammatory mass/abscess formation:

- **Fistulas** – Transmural inflammation is associated with sinus tracts that may penetrate the serosa and give rise to fistulas. Penetration of the bowel wall most often presents as an indolent process and not as the acute onset of severe abdominal pain. (See ["Overview of gastrointestinal tract perforation"](#), [section on 'Clinical features'](#).)

Fistulas are tracts or communications that connect two epithelial-lined organs. For example, fistulas may connect the intestine to colon (enterocolonic), to skin (enterocutaneous), to bowel (enteroenteric), or to the vagina (enterovaginal).

The clinical manifestation of the fistula depends upon the area of involvement adjacent to the diseased bowel segment:

- Enteroenteric fistulas may be asymptomatic or present as diarrhea because of loss of absorptive area
- Enterovesical fistulas lead to recurrent urinary tract infections, often with multiple organisms, and to pneumaturia

- Enterovaginal fistulas may present with passage of gas or feces through the vagina
- Enterocutaneous fistulas can cause bowel contents to drain to the surface of the skin

In a population-based study of 169 patients with CD, 12 patients (7 percent) had evidence of a fistula of any type at least 30 days prior to the diagnosis of CD [6].

- **Inflammatory mass/abscess** – All sinus tracts do not lead to fistulas but may present as an inflammatory mass (ie, a walled-off inflammatory mass without bacterial infection) that may be palpated on physical examination of the abdomen. Ileal involvement is suggested by a mass in the right lower quadrant.

Some sinus tracts lead to abscess formation and an acute presentation of localized peritonitis with fever, abdominal pain, and tenderness.

- **Perianal disease** – Symptoms and signs related to perianal disease may occur in approximately one-third of patients during the course of CD. For example, patients with perianal fistula may present with perianal pain and drainage, while patients with perianal abscess present with perianal pain, fever, and purulent discharge. The evaluation and management of perianal CD is discussed separately. (See "[Perianal Crohn disease](#)".)

Other gastrointestinal features — The clinical manifestations of other sites of gastrointestinal involvement in CD are variable and occur less frequently than ileocolonic involvement. Examples include:

- Oral involvement may present with aphthous ulcers or pain in the mouth and gums.
- Esophageal involvement may present with odynophagia or dysphagia.
- Gastroduodenal involvement is seen in up to 15 percent of patients and may present with upper abdominal pain, nausea, and/or postprandial vomiting [7]. The clinical features of gastroduodenal CD may be similar to those of peptic ulcer disease or gastric outlet obstruction.

The distal antrum of the stomach and duodenum are the most commonly affected upper gastrointestinal (GI) sites in patients with CD. Some patients have limited upper GI involvement and may be asymptomatic, while other patients with persistent symptoms have deep ulcerations and longer segments of intestinal involvement. (See "[Medical management of moderate to severe Crohn disease in adults](#)", section on '[Gastroduodenal disease](#)'.)

Malabsorption — Patients with small bowel CD and bile salt malabsorption may present with watery diarrhea and steatorrhea that can lead to protein calorie malnutrition, hypocalcemia, vitamin deficiency (eg, vitamin B12), and metabolic bone disease. (See ["Approach to the adult patient with suspected malabsorption"](#) and ['Extraintestinal manifestations'](#) below.)

Small bowel disease involving greater than 100 cm of terminal ileum usually results in severe impairment of the enterohepatic circulation of bile salts such that the liver's ability to upregulate de novo bile acid synthesis is inadequate to meet normal physiologic needs for bile production, resulting in fat malabsorption. (See "[Overview of nutrient absorption and etiopathogenesis of malabsorption](#)", section on '[Etiopathogenesis of malabsorption](#)'.)

Shorter segments of terminal ileal disease (ie, <100 cm) may result in chronic diarrhea, even though it may not result in fat malabsorption, since the bile acids that are not absorbed in the small intestine may stimulate water and electrolyte secretion in the colon (which is called "cholerrheic diarrhea").

Extraintestinal

manifestations — Extraintestinal manifestations of CD are generally related to inflammatory disease activity and include ([table 2](#)) [8]:

- Arthritis or arthropathy** – Primarily involving large joints in approximately 20 percent of patients without evidence of synovial destruction, this arthritis is the most common extraintestinal manifestation. Central or axial arthritis, such as sacroiliitis, or ankylosing spondylitis, may also occur. An undifferentiated spondyloarthropathy or ankylosing spondylitis may be the presenting manifestation of CD. (See "[Clinical manifestations and diagnosis of arthritis associated with inflammatory bowel disease and other gastrointestinal diseases](#)".)
- Eye involvement** – Eye manifestations occur in approximately 5 percent of patients and include uveitis, iritis, and episcleritis [9] ([picture 1A-B](#)). (See "[Dermatologic and ocular manifestations of inflammatory bowel disease](#)", section on 'Ocular disease'.)

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Table 2 - larger image below

- **Skin disorders** – Dermatologic manifestations occur in approximately 10 percent of patients and include erythema nodosum and pyoderma gangrenosum ([picture 2A-C](#)). Rarely, vulvar involvement of CD may manifest as pain, edema, erythema, and ulceration [10]. (See "[Vulvar lesions: Differential diagnosis of red lesions](#)", section on 'Crohn disease'.)

Metastatic cutaneous CD is an uncommon subtype of cutaneous disease that may affect any area but often involves the inguinal region, genitals, or oral cavity [11]. (See "[Dermatologic and ocular manifestations of inflammatory bowel disease](#)", section on 'Metastatic cutaneous Crohn disease'.)

- **Primary sclerosing cholangitis** – Primary sclerosing cholangitis typically occurs in approximately 5 percent of patients with CD, who are often asymptomatic but have an elevated serum alkaline phosphatase concentration. (See "[Primary sclerosing cholangitis in adults: Clinical manifestations and diagnosis](#)".)
- **Other hepatobiliary involvement** – Hepatobiliary disorders in patients with CD are often related to IBD medications rather than the disease itself. However, conditions such as pyogenic liver abscess are rarely seen in association with CD. (See "[Overview of hepatobiliary disorders in patients with inflammatory bowel disease](#)".)
- **Secondary amyloidosis** – Secondary amyloidosis is very rare but may lead to kidney failure and other organ system involvement [12]. (See "[AA amyloidosis: Causes and diagnosis](#)".)
- **Renal stones** – Calcium oxalate and uric acid kidney stones can result from steatorrhea and diarrhea [13]. Uric acid stones can result from dehydration and metabolic acidosis. (See "[Kidney stones in adults: Epidemiology and risk factors](#)" and "[Kidney stones in adults: Uric acid nephrolithiasis](#)".)
- **Bone loss** - Metabolic bone disease may occur as a result of glucocorticoid use and impaired vitamin D and calcium absorption [14,15]. (See "[Metabolic bone disease in inflammatory bowel disease](#)".)
- **Pulmonary involvement** – Pulmonary manifestations of IBD include bronchiectasis, chronic bronchitis, interstitial lung disease, bronchiolitis obliterans with organizing pneumonia, sarcoidosis, necrobiotic lung nodules, and pulmonary infiltrates with eosinophilia syndrome. (See "[Pulmonary complications of inflammatory bowel disease](#)", section on 'Primary respiratory involvement'.)

Physical examination — Physical examination may be normal or show nonspecific signs

(eg, weight loss) suggestive of CD. More specific findings include perianal skin tags, sinus tracts, and abdominal tenderness or palpable abdominal mass (typically in the right lower quadrant).

Laboratory findings — Routine laboratory tests may be normal or they may reveal anemia, an elevated white blood cell count, an elevated C-reactive protein, electrolyte abnormalities, iron deficiency, vitamin B12 deficiency, and vitamin D deficiency.

Stool inflammatory markers (fecal calprotectin or lactoferrin) may be elevated due to intestinal inflammation [16]. (See '[Stool inflammatory markers](#)' below.)

DIFFERENTIAL DIAGNOSIS

The differential diagnosis associated with CD is broad and varies with the site of involvement and the chronicity of the clinical presentation (eg, abdominal pain, diarrhea, weight loss):

- **Infectious colitis** – For patients with diarrhea (especially with acute symptoms), stool studies are performed to evaluate for enteric pathogens (ie, *Shigella*, *Salmonella*, *Campylobacter*, *Escherichia coli* O157:H7, *Yersinia*, *Clostridioides difficile* infection, parasites, and amebiasis. (See "[Approach to the adult with acute diarrhea in resource-abundant settings](#)".)

In immunocompromised patients, cytomegalovirus infection can mimic CD. (See "[Epidemiology, clinical manifestations, and treatment of cytomegalovirus infection in immunocompetent adults](#)", section on 'Gastrointestinal manifestations'.)

In patients with primarily small bowel involvement, *Yersinia* can cause an acute ileitis that is difficult to distinguish endoscopically from acute Crohn ileitis. Both tuberculosis and amebiasis can mimic CD of the ileum and cecum. (See "[Giardiasis: Epidemiology, clinical manifestations, and diagnosis](#)" and "[Causes of acute infectious diarrhea and other foodborne illnesses in resource-abundant settings](#)" and "[Abdominal tuberculosis](#)".)

- **Ulcerative colitis** – When CD involves the colon, it must be distinguished from UC because management of these disorders can differ. Clinical features that suggest CD include (see "[Clinical manifestations, diagnosis, and prognosis of ulcerative colitis in adults](#)", section on 'Differential diagnosis'):
 - Small bowel involvement
 - Normal appearing rectal mucosa (ie, rectal sparing) (see '[Ileocolonoscopy](#)' below)

- Stools without gross blood
- Perianal disease
- Focal or segmental disease distribution rather than continuous involvement

For some patients with IBD, the distinction between CD and UC cannot be made; such patients are referred to as having indeterminate colitis. Some patients may initially be diagnosed with UC or CD but it evolves with time to the other diagnosis. One report suggested that an evolution to a diagnosis of CD was more likely in patients initially diagnosed with UC who presented with non-bloody diarrhea or weight loss [17].

- **Diverticular colitis** – Diverticular colitis is characterized by inflammation in the interdiverticular mucosa without involvement of the diverticular orifices. In contrast, in patients with IBD and diverticulosis, the inflammation involves the colonic area harboring diverticula, as well as the diverticular orifices [18]. (See ["Segmental colitis associated with diverticulosis", section on 'Diagnosis'.](#))
- **Celiac disease** – Celiac disease is a small bowel disorder characterized by mucosal inflammation and villous atrophy which occur in response to dietary gluten. Patients with celiac disease may present with gastrointestinal symptoms that are similar to CD (eg, diarrhea, weight loss). Serologic evaluation (ie, tissue transglutaminase-IgA antibody) is typically performed to detect celiac disease in adults. The clinical manifestations and diagnosis of celiac disease are presented separately. (See ["Epidemiology, pathogenesis, and clinical manifestations of celiac disease in adults"](#) and ["Diagnosis of celiac disease in adults"](#).)
- **Irritable bowel syndrome** – Gastrointestinal symptoms such as diarrhea and abdominal pain are common to both IBS and IBD. However, patients with IBS do not have abnormal laboratory studies such as fecal calprotectin or C-reactive protein and do not have mucosal inflammation on ileocolonoscopy. (See ["Stool inflammatory markers"](#) below and ["Laboratory studies"](#) below.)
- **Lactose intolerance** – Intolerance to lactose-containing foods (primarily dairy products) is a common problem and the diagnosis may be initially confused with CD. Clinical symptoms of lactose intolerance include diarrhea, abdominal pain, and flatulence after ingestion of milk or milk-containing products. (See ["Lactose intolerance and malabsorption: Clinical manifestations, diagnosis, and management"](#).)
- **Other disorders** – Because of the segmental nature of CD, patients with a variety of disorders may have similar presenting symptoms and imaging studies. These include appendicitis, diverticulitis, ischemic colitis, and a perforating or obstructing bowel

cancer.

The clinical presentation of common variable immunodeficiency (CVID) can also resemble CD (eg, diarrhea, weight loss), and the diagnosis of CVID is discussed separately. (See ["Clinical manifestations, epidemiology, and diagnosis of common variable immunodeficiency in adults"](#), section on 'Gastrointestinal disease'.)

DIAGNOSTIC EVALUATION

Goals — Goals of the diagnostic evaluation for a patient with suspected CD are to exclude other causes for symptoms, establish the diagnosis of CD, and determine the severity of disease. (See ["Overview of the medical management of mild \(low risk\) Crohn disease in adults"](#), section on 'Assessing disease activity, severity, and risk'.)

The diagnosis of CD may be suspected in patients with compatible clinical features, including symptoms (eg, right lower quadrant abdominal pain, chronic intermittent diarrhea, fatigue, weight loss) and laboratory tests (eg, anemia, vitamin B12 deficiency, vitamin D deficiency).

The diagnosis of CD is established on the basis of radiologic, endoscopic, and/or histologic findings that demonstrate segmental and transmural inflammation of the luminal gastrointestinal tract in a patient with compatible clinical presentation (eg, abdominal pain, chronic intermittent diarrhea). Laboratory testing is complementary in assessing the severity and complications of CD, but does not establish the diagnosis.

Approach to testing — The typical workup for most patients in whom CD is suspected includes:

- Laboratory studies including blood tests and, for patients with diarrhea, stool studies. (See ["Laboratory studies"](#) below.)
- Small bowel imaging (usually computed tomography enterography (CTE) or magnetic resonance enterography [MRE] where available). (See ["Small bowel imaging"](#) below.)
- Colonoscopy with intubation of the terminal ileum, including mucosal biopsies. (See ["Ileocolonoscopy"](#) below.)

Laboratory studies — Laboratory testing is complementary and useful for assessing disease complications, while no single laboratory test can establish the diagnosis of CD [4].

Blood tests — Blood tests for patients in whom the diagnosis is suspected include:

- Complete blood count
- Blood chemistry including electrolytes, kidney function tests, liver biochemical and function tests, and blood glucose
- Serum iron, ferritin, vitamin D, and vitamin B12 levels
- Serum albumin
- C-reactive protein (CRP)

Serum CRP is an acute phase reactant that rises with inflammatory activity and is seen in some patients with IBD [19,20]. Elevated levels of CRP may help differentiate patients with IBD from those with symptoms caused by other disorders (eg, irritable bowel syndrome) [19,21-23]. CRP levels are reported to correlate with CD activity [24,25]. The use of laboratory testing for pretreatment evaluation and for monitoring patients with CD who have achieved clinical remission with medical therapy is discussed separately. (See ["Overview of the medical management of mild \(low risk\) Crohn disease in adults", section on 'Assessing disease activity, severity, and risk'](#) and ["Overview of the medical management of mild \(low risk\) Crohn disease in adults", section on 'Monitoring during remission'.](#))

Stool studies

Testing for enteric pathogens — For patients with diarrhea, a stool specimen is sent for culture, examination for ova and parasites, and *Clostridioides difficile* toxin. (See ["Clostridioides difficile infection in adults: Epidemiology, microbiology, and pathophysiology"](#) and ["Clostridioides difficile infection in adults: Clinical manifestations and diagnosis"](#).)

Stool inflammatory markers — Although tests for fecal calprotectin or lactoferrin are not used routinely to diagnose CD, they may help to differentiate patients with intestinal inflammation from patients with functional bowel disease [26-29]. While calprotectin is used more commonly by some clinicians, lactoferrin is an acceptable alternative [28,30]. (See ["Approach to the adult with chronic diarrhea in resource-abundant settings", section on 'General laboratory tests'.](#))

In selected patients (eg, those in whom it is difficult to distinguish CD from functional bowel disease), we measure fecal calprotectin instead of proceeding directly to ileocolonoscopy. If fecal calprotectin is normal, a diagnosis of IBD is unlikely. If fecal calprotectin level is above the reference range, we proceed with ileocolonoscopy and

biopsy to confirm the diagnosis of IBD.

In a meta-analysis of eight studies including 1062 participants (565 patients with IBD [confirmed by endoscopy], 259 patients with IBS and 238 healthy controls), which evaluated the performance of fecal calprotectin in distinguishing between IBD and IBS, patients with a fecal calprotectin level of <40 microg/g had a 1 percent chance or less of having IBD [28]. In a separate analysis of two studies in the same systematic review (541 participants [275 patients with IBD, 168 patients with IBS and 98 healthy controls]), patients with a fecal lactoferrin level of <10microg/g had a 2 percent chance or less of having IBD. (See "[Clinical manifestations and diagnosis of irritable bowel syndrome in adults](#)", section on 'Laboratory testing'.)

In an earlier meta-analysis that included six studies with 670 adult patients, the presence of fecal calprotectin was 93 percent sensitive and 96 percent specific for identifying patients with IBD [16], using ileocolonoscopy with histology as the reference standard.

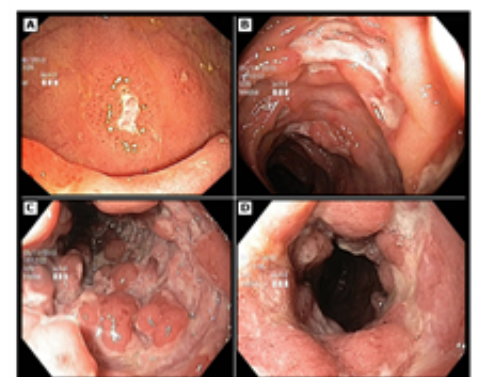
We also measure fecal calprotectin when evaluating a patient with established CD who had achieved clinical remission but who now presents with symptoms of a disease flare (eg, abdominal pain, diarrhea). In this setting, some clinicians may initiate treatment based on clinical evaluation, but we obtain objective testing (eg, stool markers, laboratory studies) prior to initiating immunosuppressive therapy.

The use of fecal calprotectin for pretreatment evaluation and for monitoring disease activity in patients with IBD is discussed separately. (See "[Overview of the medical management of mild \(low risk\) Crohn disease in adults](#)" and "[Management of Crohn disease after surgical resection](#)", section on 'Postoperative monitoring'.)

Endoscopy

Ileocolonoscopy — Colonoscopy with intubation of the terminal ileum (including mucosal biopsies) is performed for the evaluation of suspected ileocolonic CD. Endoscopic features include focal ulcerations adjacent to areas of normal appearing mucosa along with nodular mucosal changes that result in a cobblestone appearance ([picture 3](#)). (See "[Endoscopic diagnosis of inflammatory bowel disease in adults](#)".)

Endoscopic findings in Crohn disease



Endoscopic findings in Crohn

Skip areas of involvement are typical with segments of normal-appearing bowel interrupted by large areas of disease.

Pseudopolyps (hypertrophied masses of mucous membrane resembling polyps) may also be present. Normal appearing rectal mucosa (ie, rectal sparing) is common in CD. (See "[Medical management of low-risk adult patients with mild to moderate ulcerative colitis](#)", section on 'Induction of remission'.)

During colonoscopy, biopsies are obtained from the right colon, left colon, and rectum even if endoscopically normal in appearance to assess for histologic evidence of inflammation. The histologic findings include focal ulcerations and acute and chronic inflammation. These findings are usually confirmatory rather than diagnostic.

Granulomas may be noted in up to 30 percent of patients with CD and this histologic finding supports but is not required to establish the diagnosis ([picture 4](#)) [31].

Other endoscopic examinations

- **Upper endoscopy** – Upper endoscopy with biopsies of the stomach and small bowel is performed in patients with suspected CD and upper gastrointestinal symptoms (eg, dyspepsia, early satiety, nausea, emesis). Endoscopic findings consistent with CD include esophageal ulceration, gastric inflammation, and duodenal ulceration and/or inflammation.
- **Video capsule endoscopy** – Video capsule endoscopy (VCE) or wireless video endoscopy, provides an alternative method for visualizing the small bowel mucosa but without radiation exposure. It may detect suggestive lesions (eg, small bowel ulcerations) not visible by other small bowel studies ([picture 5](#)). VCE should not be performed in patients with a suspected intestinal stricture since the capsule may not be able to pass through the stricture, thus requiring surgical or endoscopic removal. (See "[Overview of deep small bowel enteroscopy](#)".)

The use and diagnostic accuracy of VCE are presented separately. (See "[Wireless video capsule endoscopy](#)".)

Imaging

Small bowel imaging — Small bowel imaging is performed as part of the diagnostic evaluation for patients with suspected CD, because much of the small bowel is not easily accessible by direct endoscopic visualization, and small bowel imaging also

disease

Picture 3 - larger image below

demonstrates disease distribution and extent. Computed tomography enterography (CTE) and magnetic resonance enterography (MRE) are routinely used to diagnose and follow CD. The imaging goal is to adequately distend the bowel to view the inner wall and timed to best evaluate bowel wall changes. CTE and MRE have similar sensitivity and specificity (sensitivity for CD ranges from 75 to 90 percent, with a specificity of >90 percent [32]) and selecting a modality includes patient variables and local expertise.

The [Society of Abdominal Radiology](#) and the [American Gastroenterological Association](#) use CTE for patients with any of the following:

- No prior cross-sectional enterography examination
- Acute illness or suspected sepsis
- Contraindication or possible safety issue related to MRI (see "[Patient evaluation for metallic or electrical implants, devices, or foreign bodies before magnetic resonance imaging](#)").

MRE is typically performed in patients who have already had a CTE, those who are <35 years of age, and pregnant patients. MRI is preferred for monitoring known CD and for assessment of perianal disease [33,34].

Enterography, both CT and MRI, requires distension of the small bowel with oral ingestion of 1.5 to 2 liters of a neutral contrast agent over the course of one hour. A neutral contrast agent is defined as one that has bowel distension capabilities with a low-density appearance similar to that of water on both CT and MRI. CT enterography is performed with intravenous iodinated contrast and with a delay of approximately 50 seconds to optimize small bowel enhancement. (See "[Prevention of recurrent reactions to iodinated contrast](#)".)

MRE includes a similar oral contrast preparation with multi-sequence, multiplanar imaging prior to and after the administration of gadolinium. (See "[Patient evaluation before gadolinium contrast administration for magnetic resonance imaging](#)".)

For patients with suspected CD, CTE is often the initial method for small bowel imaging. CTE's short examination time (5 to 10 seconds) is less affected by bowel and respiratory motion. In contrast, MRE is a longer examination (20 to 30 minutes), requires patient cooperation with breath holding, and is more susceptible to motion and respiratory artifact. MRE is excellent for monitoring disease activity given its lack of radiation, but interpretation of these examinations requires expertise [35].

CTE and MRE evaluate the entire gastrointestinal tract for wall inflammation, stricturing,

and penetrating complications. Imaging findings in CD include wall inflammation, strictures, and sequelae of penetrating disease ([image 1](#) and [image 2](#)). Inflammatory manifestations of bowel disease include segmental, asymmetric wall thickening (greater than 2 mm), wall enhancement (stratified versus homogenous), ulcerations (breaks in the intraluminal surface of the bowel), and wall edema with decreased motility (both features are MRE findings only). Stricturing disease is defined by bowel wall thickening (>3 mm), luminal narrowing (reduction of the bowel wall lumen to <50 percent of normal adjacent bowel) with upstream dilation of ≥ 2.5 cm ([image 3](#)) [36]. Upstream dilation is graded based on upstream bowel diameter. In addition, unsuccessful endoscopic passage with luminal narrowing and wall thickening at CTE or MRE also defines a stricture [37,38].

Penetrating disease complications are well seen with cross sectional imaging. A sinus tract is a blind ending tract extending beyond the bowel wall serosa but not reaching an adjacent organ. Fistulas are extra-enteric tracts connecting the bowel lumen to another epithelial surface. Fistulas may be seen as enhancing structures that may have adjacent abscesses and are named by the structures they connect ([image 4](#)). They often arise in the setting of an upstream stricture and can create a tethered appearance between the affected bowel and the other structures. An inflammatory mass is dense mesenteric inflammation without a defined fluid collection. An abscess is a fluid collection near an involved segment of bowel with or without air and can occur in the mesentery, peritoneum, retroperitoneum, and perianal soft tissues ([image 5](#)) [39].

Both CTE and MRE have the capability to image changes in the mesentery related to CD, complications of transmural disease, and extraintestinal manifestations of the disease. Mesenteric changes can be found in the distribution of segmental disease and include fibrofatty proliferation around the diseased segment of bowel ("creeping fat"), mesenteric venous occlusion, engorged vasa recta ("comb sign"), perienteric edema, and localized adenopathy [33,40].

Alternatives to CTE and MRE for small bowel imaging

Computed tomography without enteric contrast — Some patients cannot tolerate the oral preparation for enterography (eg, patients with vomiting or who are acutely ill with sepsis). We may image such patients with CT without bowel contrast but we typically use intravenous contrast for a thorough assessment of intestinal and extraintestinal complications [32].

Ultrasound — Transabdominal ultrasound for small bowel imaging is largely

dependent on operator expertise. Imaging findings include wall thickening, alterations in gut signature, fat wrapping, and hyperemia. Color Doppler ultrasound and contrast-enhanced ultrasound can help characterize strictured bowel segments [41,42]. While ultrasound has the advantage of avoiding radiation, it is highly operator dependent. Location of disease also affects utility as terminal ileal disease has higher sensitivity with ultrasound than more proximal small bowel disease. (See "[Transabdominal ultrasound of the small and large intestine](#)", section on 'Clinical applications'.)

Other diagnostic studies — Other diagnostic studies in patients with suspected CD may include:

- Small bowel enteroscopy – Small bowel enteroscopy provides direct endoscopic visualization of the small intestine in addition to allowing for tissue sampling. (See "[Overview of deep small bowel enteroscopy](#)".)
- Alternatives to colonoscopy – Colonoscopy is the preferred study for evaluation of the large bowel, while [barium](#) enema or computed tomography (CT) colonography are less desirable alternatives because the colon and ileum are not directly visualized and mucosal biopsies cannot be obtained. However, an alternative to optical colonoscopy may be indicated when it cannot be completed for technical reasons (eg, nonobstructing colonic stricture). (See "[Computed tomographic \(CT\) colonography in adults](#)".)

For example, an air-contrast [barium](#) enema may detect aphthous ulcers and can document the extent of disease and the location and severity of colonic strictures. Sacculations may be seen in patients with chronic involvement. It may also reveal small perforations with fistula tracts.

- Perianal and pelvic imaging – MRI of the pelvis is used to detect perianal fistula and define the fistula tract ([image 6](#) and [image 7](#)). The evaluation of a patient with suspected CD and perianal fistula and/or abscess is presented separately. (See "[Perianal Crohn disease](#)" and "[The role of imaging tests in the evaluation of anal abscesses and fistulas](#)".)

PROGNOSIS

Disease course — For many patients with CD, symptoms are chronic and intermittent, but the disease course can vary. Some patients may have a continuous and progressive course of active disease, while approximately 20 percent of patients experience prolonged remission after initial presentation [43,44].

Chronic bowel inflammation can lead to complications such as strictures, fistula, or abscess. A population-based cohort study showed that the risk of developing intestinal complications among patients with inflammatory CD was 50 percent at 20 years after diagnosis, and ileal involvement was associated with a shorter time interval to onset of complications [45].

Risk factors for progressive disease include [46,47]:

- Age <40 years
- Tobacco use
- Perianal or rectal involvement
- Glucocorticoid-requiring disease

Risk of surgery — Many patients with CD ultimately require surgical intervention. Although advances in medical therapy have coincided with lower rates of surgical resection in patients with CD, surgery is often required in the setting of bowel obstruction, abscesses, perforation, or refractory disease. The 10-year risk of surgical resection for CD is nearly 50 percent [48]. (See "[Surgical management of Crohn disease](#)".)

The management of patients with CD after surgical resection and the risk factors for disease recurrence are presented separately. (See "[Management of Crohn disease after surgical resection](#)".)

Cancer risk — The risk of colorectal cancer in longstanding CD involving the colon is probably comparable to UC [49-51]. However, not all studies reached these conclusions and thus the magnitude of risk in patients with CD remains unsettled. These issues, including surveillance for dysplasia, are presented separately. (See "[Surveillance and management of dysplasia in patients with inflammatory bowel disease](#)", section on 'Crohn disease'.)

Increases in incidence of squamous cell carcinoma of the anus and skin, adenocarcinoma of the small bowel, duodenal neoplasia, testicular cancer, and leukemia have all been reported in CD, although the strength of these associations is unclear [52]. In a retrospective population-based cohort study that included 3348 patients with IBD and no prior cancer history, both the overall cancer risk and the risk of colon cancer was not increased [53]. However, patients with CD had a higher risk of hematologic malignancies (standardized incidence ratios 14, 95% CI 6.7-25.9).

In addition, patients receiving thiopurine therapy for IBD had an increased risk of developing lymphoproliferative disorders [54,55]. Analyses of lymphoma risk in patients receiving biologic agents directed against tumor necrosis factor-alpha are confounded by concomitant use of immunosuppressive agents in most of these patients. The risk of lymphoma, including hepatosplenic T-cell lymphoma, is discussed separately. (See ["Overview of azathioprine and mercaptopurine use in inflammatory bowel disease"](#), section on 'Lymphoma' and ["Medical management of moderate to severe Crohn disease in adults"](#), section on 'Hepatosplenic T-cell lymphoma'.)

Mortality — Patients with CD may have a slightly higher overall mortality compared with the general population. In a meta-analysis of 35 studies, the standardized mortality ratio (an approximation of the risk of death compared with the general population) was 1.38 (95% CI 1.23-1.55) [56].

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Crohn disease in adults"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see ["Patient education: Crohn disease in adults \(The Basics\)"](#) and ["Patient education: Diet for inflammatory bowel disease in adults \(The Basics\)"](#))

- Beyond the Basics topics (see "[Patient education: Crohn disease \(Beyond the Basics\)](#)")
-

SUMMARY AND RECOMMENDATIONS

- **Background** – Inflammatory bowel disease (IBD) is comprised of two major disorders: ulcerative colitis (UC) and Crohn disease (CD). UC affects the colon and is characterized by inflammation of the mucosal layer. CD is characterized by transmural inflammation and may involve any portion of the gastrointestinal tract, from the oral cavity to the perianal area. (See '[Introduction](#)' above.)
- **Clinical features** – The cardinal symptoms of CD include crampy abdominal pain, chronic intermittent diarrhea (with or without gross bleeding), fatigue, and weight loss. The duration of symptoms suggestive of CD is variable. Some patients may present years after the onset of symptoms, while others may present acutely. (See '[Clinical features](#)' above.)

Transmural bowel inflammation of CD is associated with sinus tracts that may lead to intestinal or perianal fistulas, phlegmon, or abscess. (See '[Features of transmural inflammation](#)' above.)

Extraintestinal manifestations, such as arthritis or arthropathy, eye and skin disorders, biliary tract involvement, and kidney stones, may occur and are generally related to inflammatory disease activity ([table 2](#)). (See '[Extraintestinal manifestations](#)' above.)

- **Diagnostic evaluation** – Goals of the diagnostic evaluation of a patient with suspected CD are to exclude other causes for symptoms, establish the diagnosis of CD, and determine the severity of the disease. The diagnosis of CD may be suspected in patients with compatible clinical features including symptoms (eg, right lower quadrant abdominal pain, chronic intermittent diarrhea, fatigue, weight loss) and laboratory tests (eg, anemia, vitamin B12 deficiency, vitamin D deficiency). (See '[Goals](#)' above.)

The diagnosis of CD is established on the basis of radiologic, endoscopic, and/or histologic findings that demonstrate segmental and transmural inflammation of the luminal gastrointestinal tract in a patient with compatible clinical presentation (eg, abdominal pain, chronic intermittent diarrhea). Laboratory testing is complementary in assessing the severity and complications of CD, but does not establish the diagnosis.

The initial workup for patients in whom CD is suspected typically includes (see '[Approach to testing](#)' above):

- Laboratory studies including blood tests and, for patients with diarrhea, stool studies (see ['Laboratory studies'](#) above)
- Small bowel imaging (usually computed tomographic enterography or magnetic resonance enterography, where available) (see ['Imaging'](#) above)
- Colonoscopy with intubation of the terminal ileum, including mucosal biopsies (see ['Ileocolonoscopy'](#) above)
- **Prognosis** – For many patients with CD, symptoms are chronic and intermittent, but the disease course is variable. Some patients may have a continuous and progressive course of active disease, while approximately 20 percent of patients experience prolonged remission after initial presentation. (See ['Prognosis'](#) above.)

Many patients with CD ultimately require surgical intervention. Although advances in medical therapy have coincided with lower rates of surgical resection in patients with CD, surgery is often required in the setting of bowel obstruction, abscesses, perforation, or refractory disease. The management of patients with CD after surgical resection and the risk factors for disease recurrence are presented separately. (See ["Management of Crohn disease after surgical resection"](#).)

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Table 1: Montreal classification for Crohn disease

Age at diagnosis
A1 below 16 years
A2 between 17 and 40 years
A3 above 40 years
Location
L1 ileal
L2 colonic
L3 ileocolonic
L4 isolated upper disease*
Behavior
B1 nonstricturing, nonpenetrating
B2 stricturing
B3 penetrating
p perianal disease modifier¶

* L4 is a modifier that can be added to L1-L3 when concomitant upper gastrointestinal disease is present.

¶ "p" is added to B1-B3 when concomitant perianal disease is present.

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Table 2: Extraintestinal manifestations of inflammatory bowel disease

Common extraintestinal manifestations
Musculoskeletal
Arthritis – Colitic type, ankylosing spondylitis, isolated joint involvement such as sacroiliitis.
Hypertrophic osteoarthropathy – Clubbing, periostitis, metastatic Crohn disease.
Miscellaneous – Osteoporosis, aseptic necrosis, polymyositis, osteomalacia.
Skin and mouth
Reactive lesions – Erythema nodosum, pyoderma gangrenosum, aphthous ulcers, vesiculopustular eruption, cutaneous vasculitis, neutrophilic dermatosis, metastatic Crohn disease, epidermolysis bullosa acquisita.
Specific lesions – Fissures and fistulas, oral Crohn disease, drug rashes.
Nutritional deficiency – Acrodermatitis enteropathica (zinc), purpura (vitamins C and K), glossitis (vitamin B), hair loss and brittle nail (protein).
Associated diseases – Vitiligo, psoriasis, amyloidosis, epidermolysis bullosa acquisita.
Hepatobiliary
Specific complications – Sclerosing cholangitis (large-duct or small-duct), bile duct carcinoma, cholelithiasis.
Associated inflammation – Autoimmune chronic active hepatitis, pericholangitis, portal fibrosis and cirrhosis, granuloma in Crohn disease.
Metabolic – Fatty liver, gallstones associated with ileal Crohn disease.
Ocular
Uveitis iritis, episcleritis, scleromalacia, corneal ulcers, retinal vascular disease, retrobulbar neuritis, Crohn keratopathy.
Metabolic
Growth retardation in children and adolescents, delayed sexual maturation.
Less common extraintestinal manifestations
Blood and vascular
Anemia due to iron, folate, or vitamin B12 deficiency or autoimmune hemolytic anemia, anemia of chronic disease, thrombocytopenic purpura; leukocytosis and thrombocytosis; thrombophlebitis and thromboembolism, arteritis and arterial occlusion, polyarteritis nodosa, Takayasu arteritis, cutaneous vasculitis, anticardiolipin antibody, hyposplenism.
Renal and genitourinary tract
Urinary calculi (oxalate stones in ileal disease), local extension of Crohn disease involving

ureter or bladder, amyloidosis, drug-related nephrotoxicity.

Renal tubular damage with increased urinary excretion of various enzymes (eg, beta N-acetyl-D-glucosaminidase).

Neurologic

Up to 3% of patients may have non-iatrogenic neurologic involvement, including peripheral neuropathy, myelopathy, vestibular dysfunction, pseudotumor cerebri, myasthenia gravis, and cerebrovascular disorders. Incidence equal in ulcerative colitis and Crohn disease. These disorders usually appear 5 to 6 years after the onset of inflammatory bowel disease and are frequently associated with other extraintestinal manifestations.

Airway and parenchymal lung disease

Pulmonary fibrosis, vasculitis, bronchitis, necrobiotic nodules, acute laryngotracheitis, interstitial lung disease, sarcoidosis. Abnormal pulmonary function tests without clinical symptoms are common (up to 50% of cases).

Cardiac

Pericarditis, myocarditis, endocarditis, and heart block – More common in ulcerative colitis than in Crohn disease; cardiomyopathy, cardiac failure due to anti-TNF therapy.

Pericarditis may also occur from sulfasalazine/5-aminosalicylates.

Pancreas

Acute pancreatitis – More common in Crohn disease than in ulcerative colitis. Risk factors include 6-mercaptopurine and 5-aminosalicylate therapy, duodenal Crohn disease.

Autoimmune

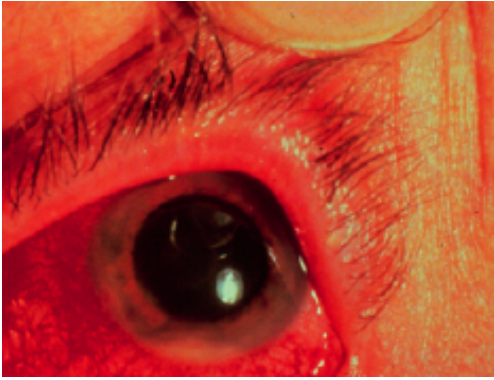
Drug-induced lupus and autoimmune diseases secondary to anti-TNF-alpha therapy.

Positive ANA, anti-double-stranded DNA, cutaneous and systemic manifestations of lupus.

TNF: tumor necrosis factor; ANA: antinuclear antibody; DNA: deoxyribonucleic acid.

Modified from: Das KM. Relationship of extraintestinal involvements in inflammatory bowel disease: New insights into autoimmune pathogenesis. Dig Dis Sci 1999; 44:1.

Picture 1A: Anterior uveitis in a patient with inflammatory bowel disease

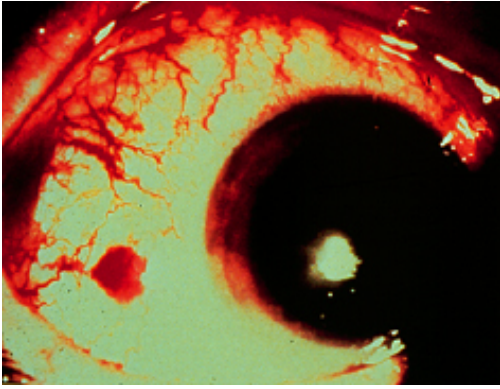


Anterior uveitis in a patient with inflammatory bowel disease is characterized by injection of the conjunctiva and opacity in the anterior chamber.

Courtesy of the American Gastroenterological Association©. This slide cannot be downloaded but may be purchased as part of a set from the AGA.

Graphic 75384 Version 5.0

Picture 1B: Episcleritis



Patient with episcleritis associated with inflammatory bowel disease showing the characteristic injection of the ciliary vessels.

Courtesy of the American Gastroenterological Association©. This slide cannot be downloaded but may be purchased as part of a set from the AGA.

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Picture 2A: Erythema nodosum



Patient with inflammatory bowel disease with red nodular areas on the shins which are characteristic of erythema nodosum.

Courtesy of the American Gastroenterological Association©. This slide cannot be downloaded but may be purchased as part of a set from the AGA.

Picture 2B: **Pyoderma gangrenosum**



Early lesion in pyoderma gangrenosum presenting as a pustular and violaceous plaque with incipient breakdown.

Courtesy of Cynthia Magro, MD.

Graphic 53733 Version 1.0

Picture 2C: Pyoderma gangrenosum

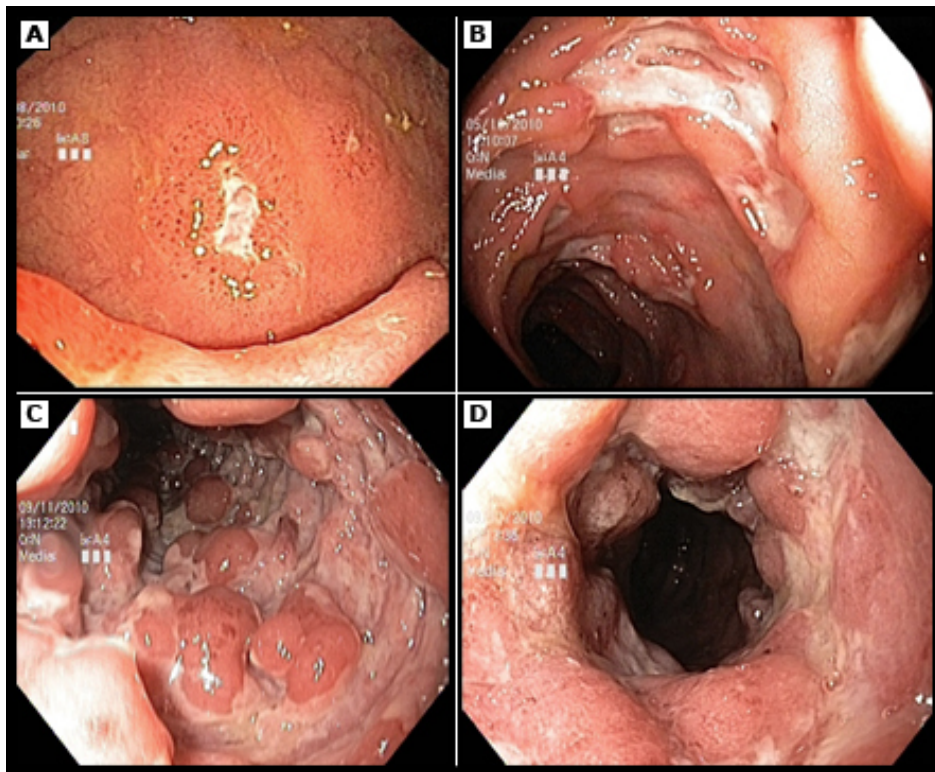


Multiple active and healing lesions of pyoderma gangrenosum with cribriform scarring in patient with inflammatory bowel disease.

Courtesy of Samuel Moschella, MD.

Graphic 52528 Version 1.0

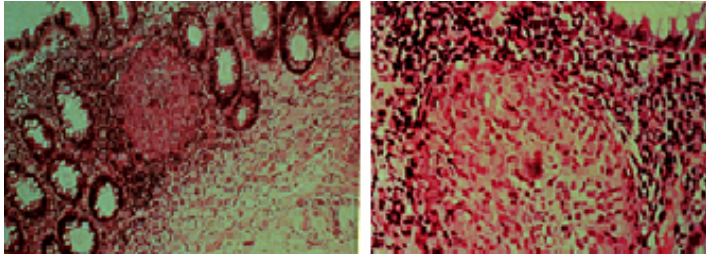
Picture 3: Endoscopic findings in Crohn disease



The dominant endoscopic feature in Crohn disease is the presence of ulcerations. Endoscopic findings in Crohn disease include: aphthous ulcers, which are the earliest lesions seen in Crohn disease (panel A); large ulcers interspersed with normal mucosa, which are typical for the segmental distribution of Crohn disease (panel B); a cobblestone appearance that is characterized by nodular thickening, with linear or serpiginous ulcers (panel C); and strictures due to fibrosis (panel D).

Courtesy of Paul Rutgeerts, MD, PhD, FRCP.

Picture 4: Typical granuloma of Crohn disease



Light micrographs showing a granulomatous lesion that is diagnostic of Crohn disease. Low and high power views show a central giant cell surrounded by epithelioid cells and rimmed by lymphocytes.

Courtesy of the American Gastroenterological Association©. This slide cannot be downloaded but may be purchased as part of a set from the AGA.

Picture 5: Crohn disease

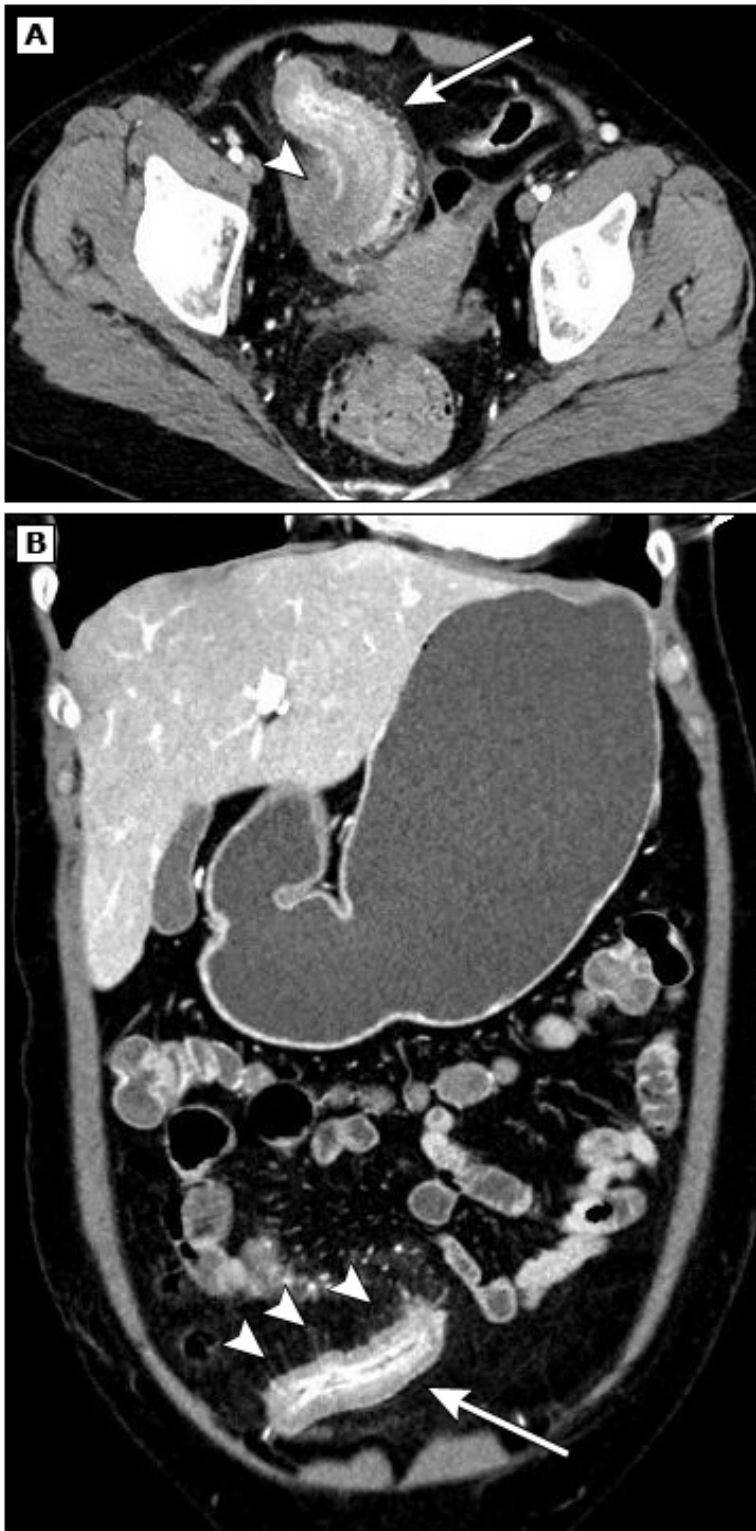


Small bowel ulceration as seen during capsule endoscopy.

Courtesy of Given Imaging, Inc.

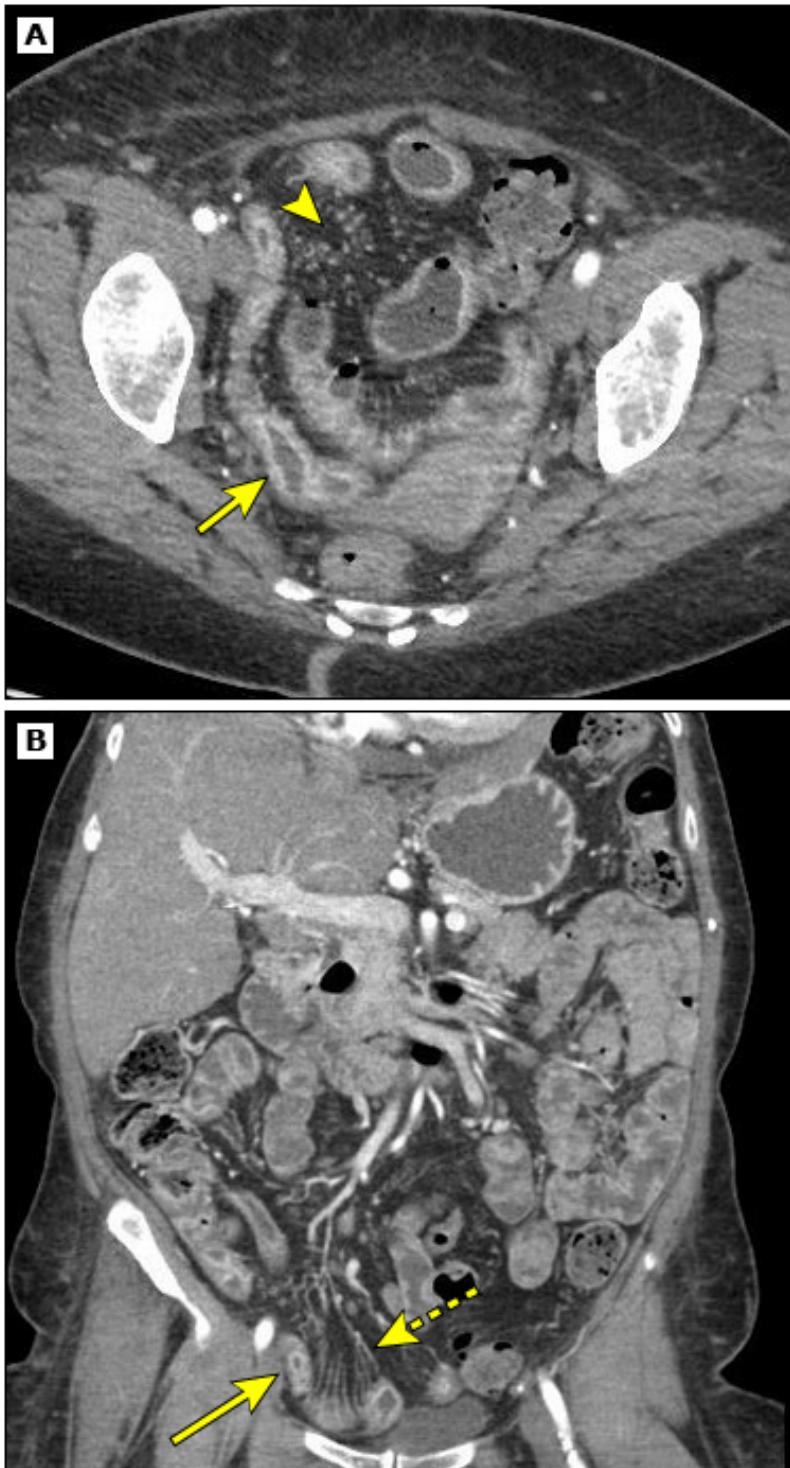
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Image 1: Ileitis related to Crohn disease on computed tomography enterography



This axial image from computed tomography enterography (A) in a patient with Crohn disease demonstrates active inflammation of a distal loop of ileum (arrow) with wall thickening and perienteric inflammation (arrowhead). The coronal image from computed tomography enterography (B) also shows active inflammation of a distal loop of ileum (arrow) with wall thickening and perienteric inflammation. Fibrofatty proliferation (arrowheads) is seen around this loop, which stands separate from the remaining bowel.

Image 2: Crohn disease with skip areas on computed tomography enterography



This axial image from computed tomography enterography (A) in a patient with Crohn disease demonstrates skip areas of inflammation of small bowel (arrow) with dilated vasa recta (arrowhead) and with wall thickening and perienteric inflammation. The coronal image from computed tomography enterography (B) also shows skip areas of inflammation of small bowel (arrow) with dilated vasa recta (dashed arrow) with wall thickening and perienteric inflammation.

Image 3: Small bowel stricturing in Crohn disease

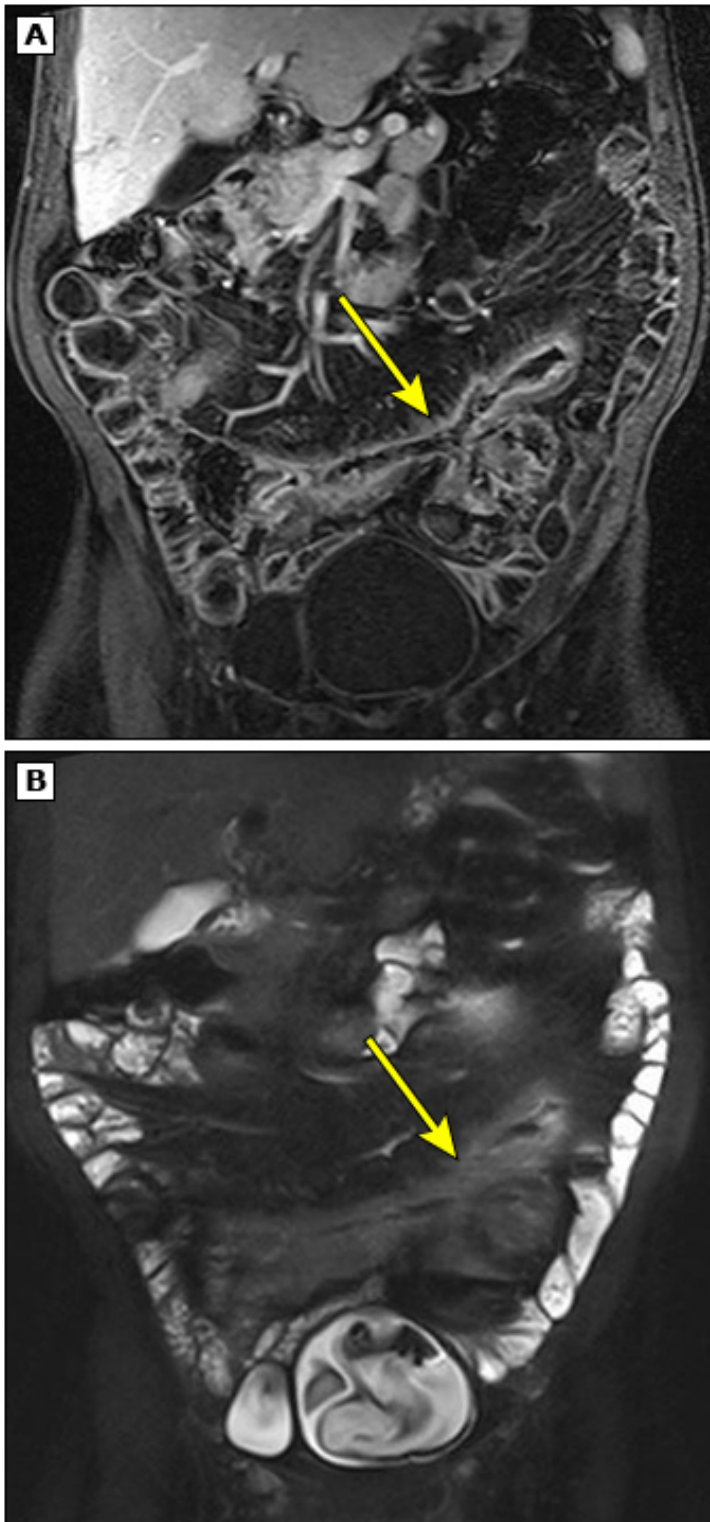


This coronal image from computed tomography enterography in a patient with Crohn disease demonstrates a strictured, inflamed segment of small bowel (arrow) causing upstream dilation and small bowel obstruction (arrowhead).

Courtesy of Tracy Jaffe, MD.

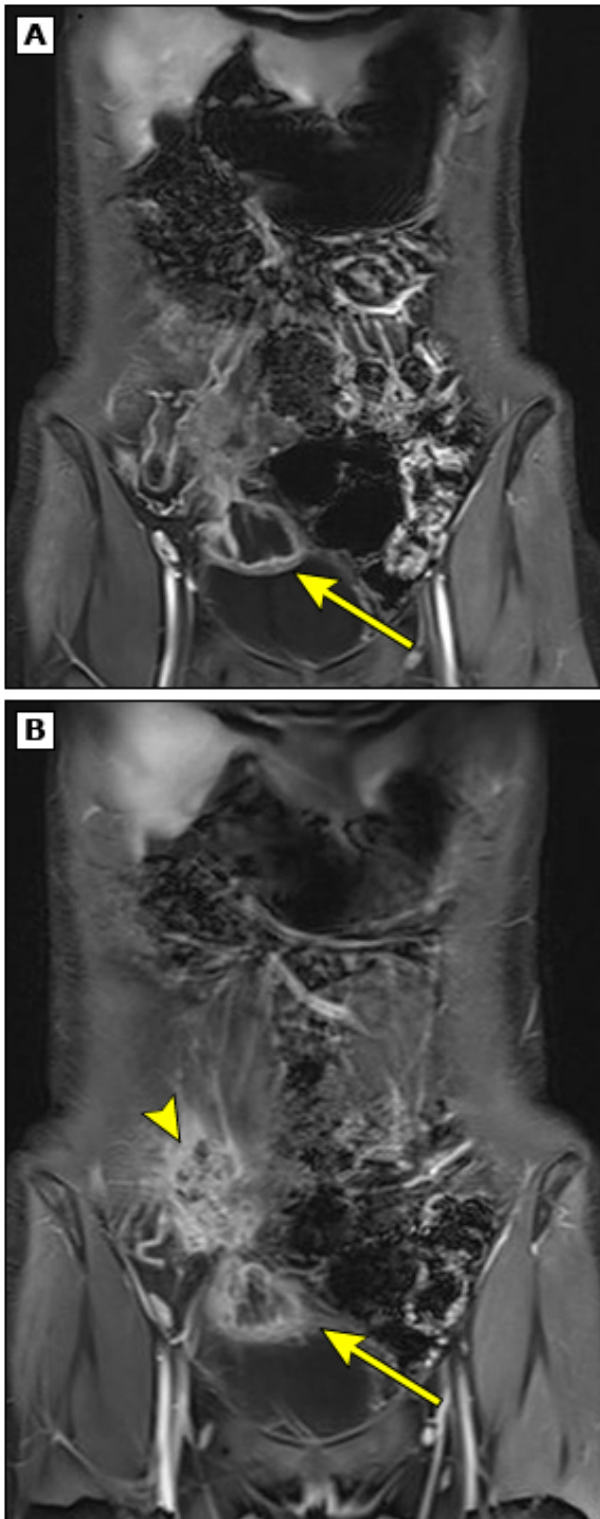
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Image 4: Fistulizing Crohn disease on magnetic resonance imaging enterography



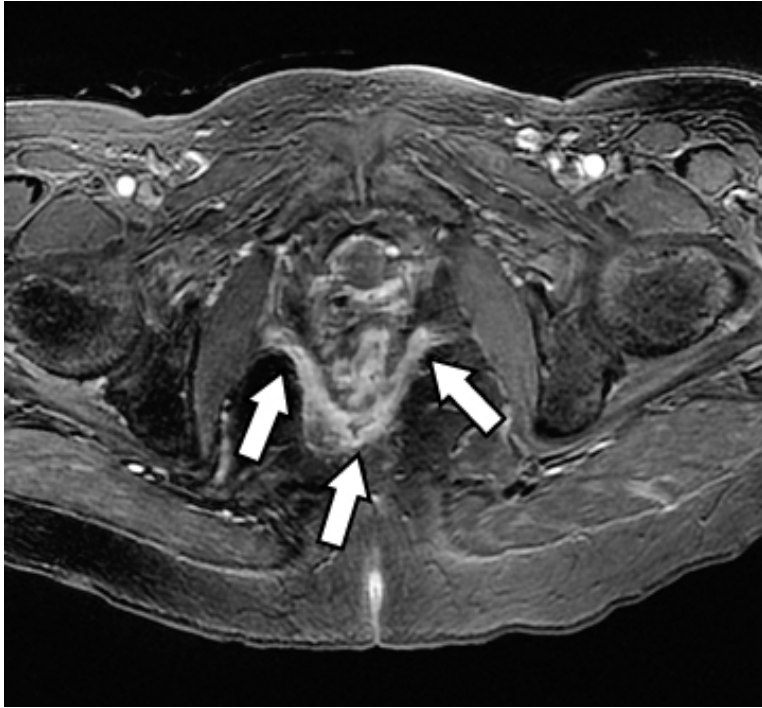
This coronal, post contrast, T1-weighted magnetic resonance imaging (MRI) study (A) demonstrates active Crohn disease with a fistula (arrow) from the small bowel to colon. The coronal T2-weighted image (B) from the MRI study also shows active Crohn disease with a fistula (arrow) from the small bowel to colon. The involved segment of bowel has increased signal consistent with bowel wall edema.

Image 5: Crohn disease with abscess on magnetic resonance imaging enterography (MRE)



This coronal, post contrast T1-weighted magnetic resonance imaging (MRI) study (A) demonstrates active Crohn disease with an abscess above the bladder (arrow). The coronal, post contrast T1-weighted image (B) also shows active Crohn disease with an abscess above the bladder (arrow) and skin (arrowhead).

Image 6: Perianal fistulas in Crohn disease

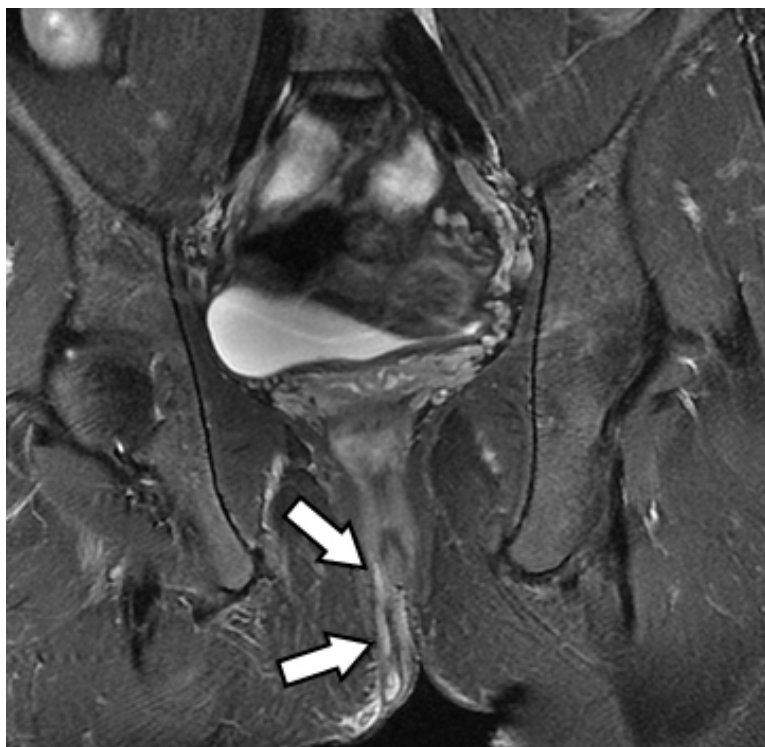


T2-weighted axial magnetic resonance imaging (MRI) study showing perianal fistulas (arrows) in a patient with Crohn disease.

Courtesy of Jonathan Kruskal, MD.

Graphic 72435 Version 5.0

Image 7: Perianal fistulas in Crohn disease



T2-weighted coronal magnetic resonance imaging (MRI) study showing a perianal fistula (arrows) in a patient with Crohn disease.

Courtesy of Jonathan Kruskal, MD.

Graphic 50476 Version 5.0

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Mark A Peppercorn, MD No relevant financial relationship(s) with ineligible companies to disclose. **Sunanda V Kane, MD, MSPH** Consultant/Advisory Boards: Boehringer Ingelheim [IBD]; Bristol Myers Squibb [IBD]; Fresenius Kabi [IBD]; InveniAI [IBD]; Janssen [IBD]; Lilly [IBD]; Takeda [IBD]. Other Financial Interest: PredicaMed [Scientific Board]. All of the relevant financial relationships listed have been mitigated. **Jana Al Hashash, MD, MSc, FACP, AGAF** Consultant/Advisory Boards: Bristol Myers Squibb [Ulcerative colitis]. All of the relevant financial relationships listed have been mitigated. **Tracy Jaffe, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Kristen M Robson, MD, MBA, FACP** No relevant financial relationship(s) with ineligible companies to disclose.

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